P07 | Contribution, evidence and limits

For David Celeita and Masoud Barati | Review draft | 03 Nov 2026, 12:46 Colombia

Contribution

A sparse-PMU procedure compares new load/line interventions with the response retained after restoration. Switched DAE predictors start from estimated past states; active and originating-cause weights distinguish an intervention from its continuing effect. This is an electrical history-filter construction, not a new Bayes rule, IMM/M3H recursion or tensor factorization.

Independent simulated comparison

Procedure	Timely family + source	Extra alarms
Physical bank	644/648	72
Nominal bank	643/648	92
ExtraTrees	330/648	29
ExtraTrees short	352/648	16

108 interventions replayed under three network conditions and two noise levels; 18 event parents, not 648 independent experiments. Unequal alarm operating points.

What the controlled results establish

On 18 existing continuous parents, preserving physical-state deviations gives 108/108 timely attributions and 1 extra; resetting them gives 75/108 and 166 extras. Marginal and bounded-profile scores tie on those endpoints. This posthoc test supports state carry, not a general necessity claim for Bayes. A development-only continuous single-state GLR control exists; it still needs independent validation under matched information and calibration.

Difference from the hackathon and the literature

The gain over the fixed causal ExtraTrees adapters is predominantly timely load coverage; line coverage is similar to the short-window model. This is not an improved official hackathon score. Physical trajectory diagnosis and M3H histories have direct precedents. The useful result is recovery-consistent attribution with explicit operational errors and limitations, rather than novelty of the individual ingredients.

Corrections and remaining submission blockers

The manuscript now uses monitored exposure (108 s per 120-s event record); alarm-rate differences and their intervals rise by 120/108, without changing predictions or matches. It states the balanced phase-A/47-coordinate map, observation memory and measured compute cost. Remaining scientific gaps: independent continuous-observer/control comparison at matched information and alarm budgets, cross-model transfer, and calibration of attribution masses. Author approval and declarations remain open. B01 is separate and still running; no partial outcome is used here.